

PROJECT REPORT

InternFlow

AI-Powered Internship Recruitment SaaS — Project Report

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Repository: <https://github.com/MohanPrabu018-K/Internflow>
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1. Executive Summary

InternFlow is a production-ready, enterprise-grade web application that manages the complete internship recruitment lifecycle — from application intake to onboarding. It is built with Next.js 15 (App Router), React 19, TypeScript (strict mode), Tailwind CSS, Prisma ORM with PostgreSQL, NextAuth.js v5 authentication with role-based access control, and more than ten AI services supporting screening, ranking, assessment generation, and recruiter assistance.

2. Problem Statement

Internship recruitment typically runs across disconnected spreadsheets, inboxes, and interview notes. Recruiters lack a single pipeline view, candidate evaluation is inconsistent, and onboarding checklists are tracked manually. InternFlow addresses this by unifying intake, screening, assessment, interviews, offers, onboarding, and analytics in one governed system.

3. Objectives

- Provide a complete Kanban recruitment pipeline (New to Joined) with drag-and-drop movement.
- Centralise candidate data, resumes, assessments, interviews, offers, and communication.
- Enforce role-based access (Admin / Recruiter / Viewer) with secure authentication.
- Embed AI assistance for screening, ranking, assessment generation, and recruiter queries.
- Deliver analytics on funnel, departments, colleges, and monthly trends.

4. System Architecture

The system is a Next.js full-stack application: server-rendered pages and API routes in the App Router, a Prisma data-access layer over PostgreSQL, NextAuth.js middleware for auth gating and rate limiting, and service modules for AI, email, storage, and security.

Layer	Implementation
Presentation	Next.js 15 App Router — 22 routes; React 19 + Tailwind CSS 3 + Recharts dashboards
API	Next.js route handlers (auth, health, applications, candidates, analytics)
Auth	NextAuth.js v5 — Google OAuth + credentials (bcrypt), JWT sessions, RBAC middleware
Data	PostgreSQL + Prisma 7 — 15 models, migrations, seed script (100 candidates)
Services	10 AI services + 14 API services; Resend email; local/S3-ready storage
Quality	TypeScript strict, ESLint, Zod validation, Vitest tests

5. Key Features

- 7-stage Kanban pipeline with drag-and-drop and candidate data tables (search, sort, filter, bulk actions).
- Candidate profiles with resume, assessments, timeline, and AI analysis tabs.
- Assessment templates with an AI question/coding-challenge generator and live coding evaluation.
- Interview scheduling, 8 offer-letter templates with PDF generation, and a 7-template email center.
- Analytics dashboard: funnel chart, department distribution, college-wise stats, monthly trends.

- 10 AI services: resume screening/parsing, skill badges, recruiter copilot, ranking leaderboard, duplicate detection, JD generator, chatbot.

6. Data Flow

- Application intake → validation (Zod) → PostgreSQL via Prisma → candidate pipeline.
- Screening → AI match scoring → assessment/interview stages → offer → onboarding checklist.
- All state transitions are auth-guarded; analytics aggregate from the same models.

7. Implementation Highlights

- Role-gated pages and API routes; 30-day JWT sessions; rate limiting, XSS sanitisation, security headers, CSRF protection.
- Health-check endpoint (/api/health) with database connectivity test.
- Seed script provisions departments, colleges, users, and 100 candidates for demo.

8. Testing & Verification

- Production build: 22 routes, 0 TypeScript errors, 0 ESLint errors, 102 kB shared first-load JS.
- Vitest suite (7 tests passing); type-check and lint scripts in CI-style workflow.

9. Results

The repository builds cleanly with zero type and lint errors and delivers a complete, navigable recruitment product: landing site, onboarding wizard, dashboard, pipeline, assessments, interviews, offers, emails, reports, settings, candidate portal, and ranking leaderboard.

10. Challenges & Future Scope

- Challenge: unifying 17 feature modules and 30+ components under strict type safety — addressed with shared types, Zod schemas, and service abstractions.
- Future: hosted production deployment (guide exists in DEPLOYMENT.md), S3 storage cutover, deeper analytics, and expanded AI evaluation.

11. Repository & Reproduction

Repository: <https://github.com/MohanPrabu018-K/Internflow> — clone, configure .env.local (PostgreSQL, NextAuth secret, optional Google OAuth/Resend keys), run npm install followed by npm run dev, and open <http://localhost:3000>. Seeded demo accounts for Admin, Recruiter, and Viewer roles are documented in the repository README.